

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 3 – Article Review (5 QUESTIONS)

Course Code:	MLA0407	Course Title:	Deep Learning
CO Assessed:	CO4	Assessment:	Assessment 3: Article Review
Weightage:	35% of CIA	Total Marks:	(5 Questions × 10 Marks)
CO1 Statement:	<i>Apply the fundamental concepts of machine learning and neural networks to design and analyze learning algorithms using perceptrons, multilayer perceptrons, forward propagation, backpropagation, gradient descent optimization techniques, and Maximum Likelihood Estimation for solving real-world problems.</i>		
Student Name:	CH. Durga Prasad	Reg. No.:	192425305
Section / Batch:	AIML(2024)	Date:	29.08.2026

Code of Conduct

I CH. Durga prasad (1924253025) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: CH. Durga Prasad

QUESTIONS: 5

Total Marks: 50

Annexure A

Article Review

1. Transfer Learning in Medical Image Analysis

Review a recent research article on the application of Transfer Learning for medical image classification. Critically analyze the problem addressed, dataset used, pre-trained model, transfer learning technique, results, limitations, and possible improvements.

2. Fine-Tuning Deep Learning Models

Review a research article that investigates fine-tuning strategies for pre-trained deep learning models. Analyze how different layers are fine-tuned and discuss their impact on model accuracy and generalization.

3. DenseNet-Based Image Classification

Review a research article that uses DenseNet for an image classification problem. Explain the DenseNet architecture, dense connections, methodology, experimental results, advantages, limitations, and future scope.

4. Transfer Learning for Industrial Applications

Review a research article that applies Transfer Learning to solve an industry-related problem, such as defect detection, agriculture, healthcare, autonomous vehicles, or manufacturing.

5. RNN for Sequential Data Processing

Review a research article that applies Recurrent Neural Networks (RNNs) to a sequential data problem. Analyze the architecture, sequence representation, training process, BPTT mechanism, results, and limitations.

6. LSTM for Sentiment Analysis

Review a research article that uses Long Short-Term Memory (LSTM) networks for sentiment analysis or text classification. Critically examine how LSTM handles long-term dependencies compared with traditional RNNs.

7. Bidirectional RNN for Natural Language Processing

Review a research article that uses Bidirectional RNN (BiRNN) or Bidirectional LSTM for an NLP application. Analyze the benefits of forward and backward sequence processing and evaluate the reported results.

8. Encoder–Decoder Architecture

Review a research article based on the Encoder–Decoder architecture for an application such as machine translation, text summarization, speech recognition, or conversational AI. Explain the architecture and critically evaluate its effectiveness.

9. Sequence-to-Sequence Learning

Review a research article that implements a Sequence-to-Sequence (Seq2Seq) model. Analyze the input-output sequence mapping, model architecture, training methodology, evaluation metrics, and research findings.

10. Comparative Review of Deep Learning Architectures

Review a research article that compares two or more architectures, such as RNN vs LSTM, LSTM vs BiLSTM, or Transfer Learning vs training from scratch. Critically analyze the experimental setup and justify whether the conclusions are supported by the results.

ARTICLE REVIEW – 1

Transfer Learning in Medical Image Analysis

Article

“Precision meets generalization: Enhancing brain tumor classification via pretrained DenseNet with global average pooling and hyperparameter tuning”

1. Problem Addressed

The study focuses on automatically classifying brain tumors from MRI images. Manual diagnosis can be difficult because tumors differ in size and appearance.

2. Dataset

The researchers used the **Figshare brain tumor dataset**, containing **3,064 T1-weighted contrast-enhanced MRI images from 233 patients**. The dataset contains three tumor types: meningioma, glioma and pituitary tumor.

3. Model Used

The study compared four pre-trained models:

- ResNet
- EfficientNet
- MobileNet
- DenseNet

The models used **ImageNet transfer learning**.

4. Transfer Learning Technique

MRI images were preprocessed and given to a pre-trained DenseNet model for feature extraction. Global Average Pooling and classification layers were then added to classify the tumor type.

5. Results

DenseNet achieved the highest test accuracy of **96%**, compared with **91% for ResNet, 91% for EfficientNet and 93% for MobileNet**.

6. Limitations

The dataset is relatively limited in size and comes from a specific source, so performance may change on images from other hospitals or scanners.

7. Possible Improvements

- Use larger multi-hospital datasets.
- Apply stronger data augmentation.
- Use attention mechanisms.
- Perform external clinical validation.
- Add explainable AI methods.

Conclusion

The article demonstrates that transfer learning with DenseNet can achieve high accuracy while reducing the need to train a deep network from scratch.

ARTICLE REVIEW – 2

DenseNet-Based Image Classification

Article

“Classification of asymmetry in mammography via the DenseNet convolutional neural network” (2023).

1. Problem Addressed

The study applies DenseNet to classify **asymmetry in mammography images**, supporting automated breast-image analysis.

2. DenseNet Architecture

DenseNet connects each layer to subsequent layers so that feature information can be reused. The network contains convolutional layers, dense blocks, transition layers and a final classification layer.

3. Dense Connections

Each layer receives information from earlier layers. This improves feature reuse and helps information flow through the network.

4. Methodology

The researchers trained a DenseNet-based CNN to identify mammographic asymmetry. The approach uses deep image features rather than relying on manually designed features.

5. Advantages

- Feature reuse
- Better information flow
- Fewer redundant parameters
- Helps gradient propagation
- Effective image feature extraction

6. Limitations

DenseNet models can require considerable computational resources, and medical-image datasets may not fully represent the diversity encountered in clinical practice.

7. Future Scope

- Larger datasets
- Multi-center validation
- Explainable AI
- Combining DenseNet with attention mechanisms
- Real-world clinical testing

Conclusion

DenseNet is suitable for medical image classification because dense connections encourage feature reuse and effective information propagation.

ARTICLE REVIEW – 3

RNN for Sequential Data Processing

Article

“Gradient-free training of recurrent neural networks using random perturbations” (2024).

1. Problem Addressed

The article investigates training difficulties associated with recurrent neural networks, particularly the computational challenges of **Backpropagation Through Time (BPTT)**.

2. RNN Architecture

The RNN processes sequential input step by step. Each hidden state carries information from the previous time step and combines it with the current input. **3. Sequence Representation** For a sequence:

$x_1, x_2, x_3, \dots, x_t$ the hidden state is

represented as: $h_t = f(x_t, h_{t-1})$

Thus, the current state depends on both the current input and the previous hidden state.

4. Training Using BPTT

BPTT unfolds the RNN across time and propagates the error backward through the entire sequence to update the network parameters.

5. Problems with BPTT

BPTT requires storing gradient information and alternating between forward and backward phases. It can also suffer from vanishing gradients when processing long sequences.

6. Proposed Direction

The article investigates **gradient-free training using random perturbations** as an alternative to conventional BPTT.

7. Limitations

Gradient-free approaches may require additional computational sampling and may not perform better than BPTT for every sequence-learning problem.

Conclusion

The article highlights an important limitation of traditional RNN training and explores an alternative method to reduce dependence on BPTT.

ARTICLE REVIEW – 4

LSTM for Sentiment Analysis Article

“Sentiment Analysis of Covid-19 Twitter Data using Deep Learning Algorithm” (2024).

1. Problem Addressed

The research analyzes public sentiment in Twitter data related to COVID-19 using an LSTMbased deep learning approach.

2. Dataset

The study used **179,107 COVID-19-related tweets**.

3. Preprocessing

The tweets were cleaned before training. Text preprocessing was performed using the **Neat Text** module.

4. Model Architecture

The processed tweets were tokenized and converted into numerical representations using word embeddings. These embeddings were given to an LSTM layer followed by dense layers for sentiment classification.

5. Why LSTM?

LSTM contains memory cells and gates that help preserve important information over longer sequences. Compared with a traditional RNN, it is better at handling long-term dependencies.

6. Results

The reported LSTM model achieved approximately **96% accuracy**, while the compared ANN approach achieved approximately **76% accuracy**.

7. Limitations

Twitter language contains slang, abbreviations, sarcasm and rapidly changing expressions, which can make sentiment classification difficult.

8. Improvements

- Use larger and more diverse datasets.
- Handle sarcasm explicitly.
- Use attention mechanisms.
- Compare against Transformer models.
- Perform multilingual sentiment analysis.

Conclusion

The study demonstrates that LSTM can effectively capture sequential information and achieve strong performance in sentiment analysis.

ARTICLE REVIEW – 5

Encoder–Decoder Architecture

Article

“In-Image Neural Machine Translation with Segmented Pixel Sequence-to-Sequence

Model” (2023).

1. Problem Addressed

The research addresses **translation of text contained inside images**. Traditional systems usually use OCR followed by machine translation and text rendering.

2. Proposed Approach

The proposed method uses an end-to-end encoder-decoder architecture that directly processes segmented pixel sequences instead of depending on a separate OCR and translation pipeline.

3. Encoder–Decoder Architecture

The encoder processes the input image and converts it into an encoded representation. The decoder uses this representation to generate the translated output.

4. Training

The model was trained end-to-end using a dataset containing **4 million English-German image pairs**.

5. Results

The proposed approach reported:

- Higher translation quality than the cascade approach.
- Better robustness to out-of-domain data.
- Less sensitivity to incomplete words.
- Reduced error propagation between OCR and translation.

6. Advantages

- End-to-end processing
- Avoids separate OCR errors
- Better handling of incomplete words
- Robust to some out-of-domain cases

7. Limitations

The model requires a large paired dataset and significant computational resources. It may also face difficulties with unusual fonts, complex layouts and low-quality images.

8. Future Scope

- Larger multilingual datasets
- Better handling of complex layouts
- More efficient architectures
- Real-time image translation

Conclusion

The article demonstrates that an encoder-decoder architecture can perform image-based translation end-to-end and reduce error propagation compared with traditional OCR-plustranslation pipelines.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, wellanalyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
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